EXPERIMENT 1

Read an image in Python and convert it to grayscale.

Aim

Read an image in Python and convert it to grayscale.

Algorithm

• Import OpenCV and read the input image.

• Check that the image was loaded successfully.

• Convert the image from BGR to grayscale.

• Save the grayscale image.

• Display the grayscale result.

Code

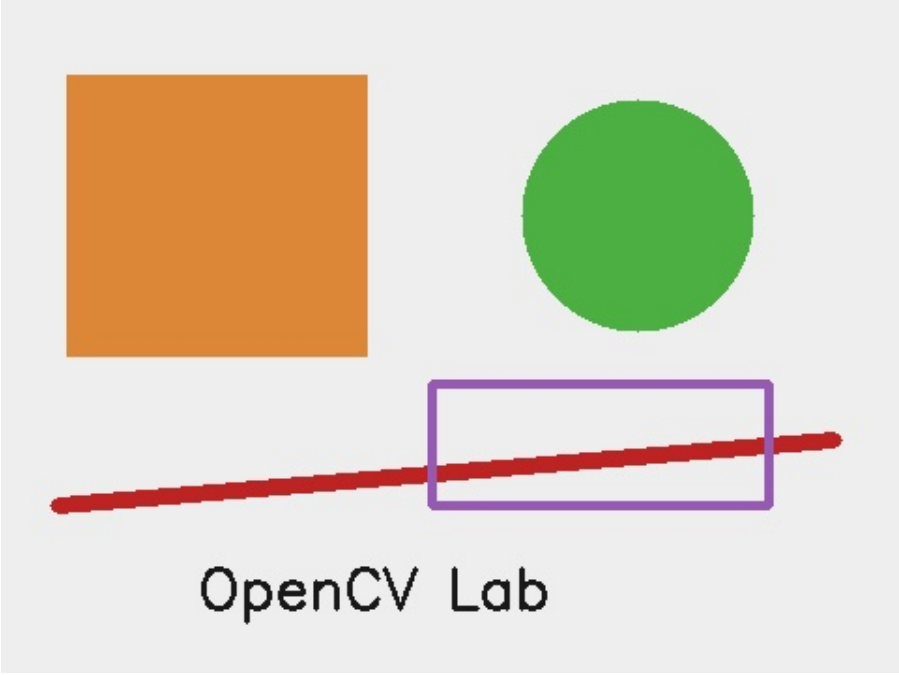
```
import cv2

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2.imwrite("output_images/experiment01_grayscale.png", gray)
print("Saved output_images/experiment01_grayscale.png")
```

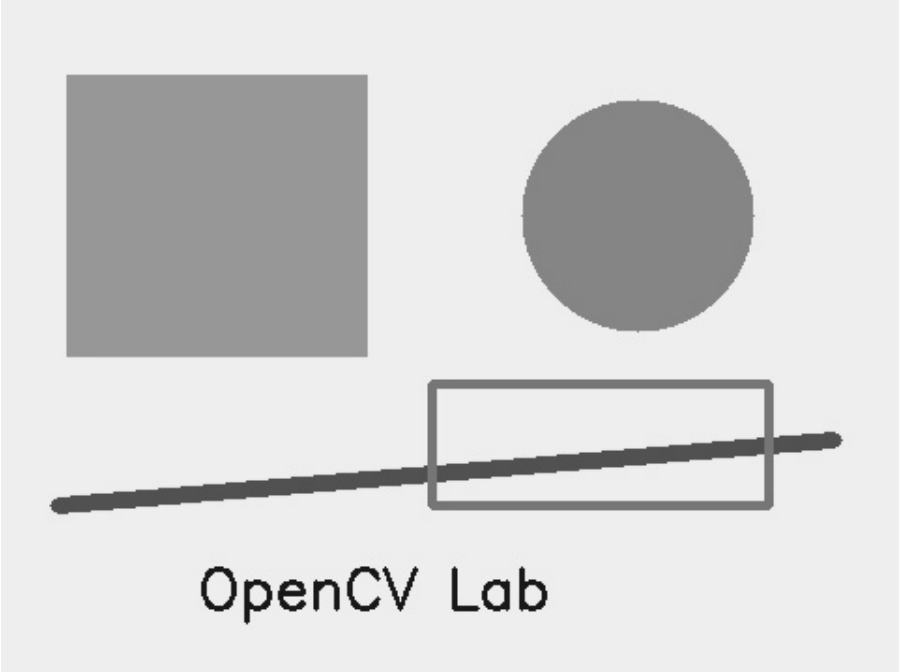
Input Image:

Output Image:

Input Image



Output Image


OpenCV Lab